

TETR: Traversability Evaluation with Transformers For Autonomous Navigation in 3D Terrain Environments

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Abstract—Autonomous navigation of unmanned ground vehicles (UGVs) in complex three-dimensional terrains necessitates precise evaluation of road traversability, a challenge that remains unresolved due to the difficulty in quantifying passable regions. Existing methodologies often fall short in effectively evaluating traversability in complex environments. In this paper, we present a novel approach for evaluating terrain traversability by enabling the model to implicitly learn principle-based knowledge. The approach utilizes elevation maps obtained from equipped sensors of UGVs, capturing detailed terrain information through a novel transformer-based model, enabling accurate and efficient traversability assessments. We evaluated our method using a comprehensive simulation dataset, which achieves an accuracy of 93.1%, representing a 5.2% improvement over existing baseline method. Furthermore, our approach demonstrated a 16.9% reduction in computational overhead, facilitating real-time implementation.

Index Terms—Autonomous Navigation, Traversability Evaluation, Transformer, Self-Supervised Learning

I. INTRODUCTION

Autonomous navigation in complex rough terrains, such as off-road environments, poses significant challenges for UGVs [1]. Terrain traversability evaluation evaluates whether a region is passable and safe for UGVs to navigate without encountering hazards. Thus, the ability to accurately assess the terrain traversability is crucial for autonomous navigation in complex three-dimensional terrains. Enabling UGVs to handle this capability remains a complex task, many research teams have conducted significant work on this problem.

Navigating in challenging terrain may increase the risk of autonomous navigation, causes material stress, leads to passenger discomfort, and demands more energy. Prior studies generally represent terrain traversability using grid-based cost maps. These cost maps are utilized by path planners to determine the optimal route from UGVs' current position

to its target position [2]. The cost maps are represented binarily, which simply classify the terrains of surrounding environment as traversable or non-traversable areas [3]. This approach suffice in simple scenarios, for example, scenarios that include clearly defined impassable obstacles. However, real-world scenarios often include regions that are passable but entail significant costs [4]. Hence, terrain traversability should be characterized by a continuous cost map which considers multiple metrics such as inclination.

Recent approaches utilize learning-based methods to achieve a more thorough understanding of the scene, which can be summarized by their network architectures and learning paradigms. Wellhausen et al. [5] utilized manually labeling data to train a CNN for predicting suitable terrain from elevation maps. Similarly, Chavez-Garcia et al. [6] utilized hand-labeled traversability knowledge obtained through the vehicle's quasi-static roll model, thereby improving the safety and efficiency of wheeled robot navigation [7]. These studies underscore the importance of high-quality, annotated datasets in providing clear supervision signals that enable models to learn intricate terrain characteristics essential for robust locomotion and navigation in diverse environments. Supervised learning approaches face significant challenges due to the high cost and effort required to obtain labeled datasets [8]. CNN inherently focus on local feature extraction through convolutional kernels with fixed receptive fields, which falls short on capturing global contextual information and long-range dependencies [9]. In this paper, we present a novel approach that enables the model to implicitly learn principle-based information. Leveraging the elevation map, our method captures intricate details of the terrain and processes spatial relationships more effectively than traditional convolutional neural networks. The transformer's attention mechanism al-

lows the model to focus on relevant features in the data, enhancing the accuracy of traversability assessments. The main contributions of this work are list as follows:

- A novel transformer-based model for traversability evaluation is proposed, enhancing both the accuracy and computational efficiency significantly compared to existing approaches.
- An innovative principle-based encoding process is proposed that enables the model to implicitly learn principle-based information, which has been empirically validated to improve the model's fitting capabilities.
- A self-supervised training framework is developed, which utilizes a physics engine that alleviates the reliance on costly manually labeled datasets, enabling scalable model training.

II. APPROACH

Evaluating terrain traversability is essential for the effective operation of UGVs in terrain environments. In this section, we define the traversability evaluation problem and introduce the proposed method. TETR leverages an encoder-only transformer architecture to analyze elevation voxel maps, integrating vehicle dynamics with terrain features to produce accurate traversability maps. Finally, we detail the training procedures and optimization strategies used to enhance the model's performance and reliability.

A. Problem Definition

Given a representation of the surrounding environment modeled using an elevation voxel map $M(x, y)$ with a resolution of d_{voxel} . Here, $x \in [0, W]$ and $y \in [0, H]$ represent the coordinates in the lateral and longitudinal directions, respectively, where W and H denote the width and height of the elevation map [10].

The goal of traversability evaluation is to verify whether the UGV can operate on a specific terrain. Finding the function f_{eval} which predicts each element within the traversability cost value, which is represented as $C \in \mathbb{R}^{H \times W}$:

$$C = f_{\text{eval}}(M) \quad (1)$$

B. TETR: Traversability Evaluation Transformer Network

In this section, an only-encoder transformer architecture is established to predict the traversability of the given elevation map. An overview of the model is depicted in Fig. 1.

1) *Principle-Based Encoding*: The UGV's traversability is largely related to attitude changes caused by the terrain. We integrate the vehicle's quasi-static roll model into the processing of the elevation map to enhance the traversability assessment. For each voxel within the elevation map, we uniformly sample the vehicle's heading angles at intervals of 10° , covering the full range from 0° to 360° . For each voxel at position (x_i, y_i) , we uniformly sample the vehicle's heading angles θ_i at intervals of 10° , covering the full range from 0°

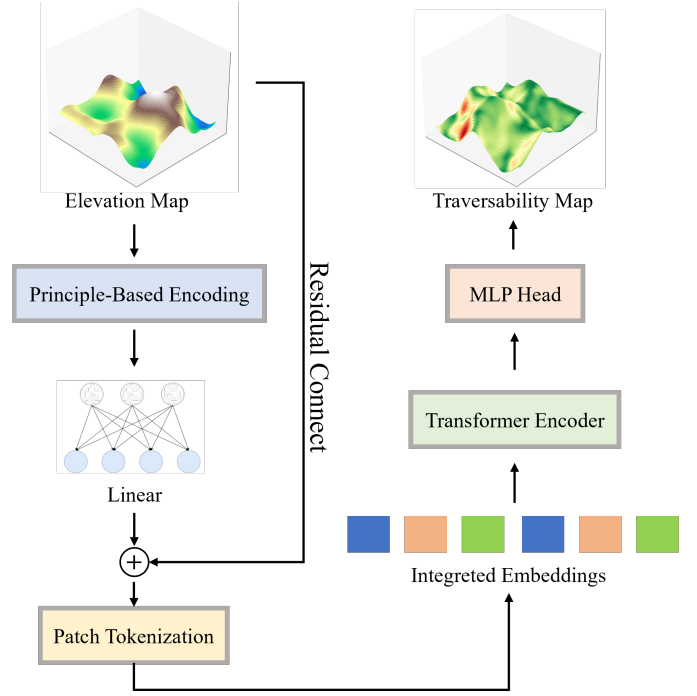


Fig. 1. The overall architecture of the proposed TETR method. The input of TETR is the elevation map obtained from sensors equipped on UGVs, after the inferring process, the traversability map is evaluated as the model's output.

to 360° . The corresponding roll angles φ_i of the UGV can be extracted from the elevation map as:

$$\varphi_i = \arcsin \left(\frac{M(x_1, y_1) - M(x_2, y_2)}{L_{xy}} \right) \quad (2)$$

where L_{xy} , $M(x_1, y_1)$, and $M(x_2, y_2)$ are defined as:

$$\begin{cases} L_{xy} = L \cos \phi \\ M(x_1, y_1) = M(x_i + L_{xy} \cos \theta_2, y_i + L_{xy} \sin \theta_2) \\ M(x_2, y_2) = M(x_i - L_{xy} \cos \theta_2, y_i - L_{xy} \sin \theta_2) \end{cases} \quad (3)$$

where L is the track width of the UGV, L_{xy} is the projection length of the wheelbase of the UGV on the specific terrain, $M(x_1, y_1)$ and $M(x_2, y_2)$ are the elevation values of the voxel.

Leveraging the vehicle's quasi-static roll model, the maximum allowable lateral acceleration before rollover is:

$$a_{latmax} = \left(\frac{L}{2h_g} + \varphi_i \right) g \quad (4)$$

where h_g is the height of the centroid and a_{latmax} is the lateral acceleration limitations.

Building upon the previously described methodology, the computed maximum lateral acceleration a_{latmax} is systematically integrated with the corresponding heading angles θ_i and positional information derived from the elevation map to construct a three-dimensional tensor, which serves as a principled encoding method, effectively capturing the intricate relationships between vehicle dynamics and terrain characteristics. The tensor T can be expressed as:

$$T(x_i, y_i, \theta_i) = a_{latmax} \quad (5)$$

By structuring the data into this three-dimensional tensor, the transformer-based model is equipped with a comprehensive representation that encapsulates both the geometric features of the terrain and the dynamic constraints of the vehicle. To ensure alignment with the two-dimensional elevation map features, a linear layer is applied to the three-dimensional tensor $T(x, y, \theta)$, which encapsulates the quasi-static rollover model information. This operation reduces the dimensionality of the tensor, producing a two-dimensional feature map (x,y). The transformation is mathematically expressed as follows:

$$T'(x, y) = \text{Linear}(T(x, y, \theta)) \quad (6)$$

To preserve the intrinsic elevation information within the terrain features, we implement a residual connection that integrates T' with the original elevation map features [11]. This residual connection ensures that the essential geometric characteristics of the terrain are retained, thereby enhancing the model's ability to accurately assess traversability. The combined feature representation serves as the initial input to the Transformer Encoder, facilitating the effective fusion of dynamic vehicle stability data with static terrain information. Formally, the residual connection is defined as:

$$F_{\text{integ}} = T' + M \quad (7)$$

where F_{input} is the integrated feature map serving as the input to the Transformer Encoder.

2) *Patch-Based Embedding Generation:* The standard Transformer receives as input a one-dimensional sequence of token embeddings. Inspired by vision transformer [12], to effectively leverage the rich spatial information inherent in the elevation map, we reshape the input feature $F_{\text{integ}} \in \mathbb{R}^{H \times W}$ into a sequence of flattened two-dimensional patches $P \in \mathbb{R}^{N \times p^2}$, where (H, W) represents the resolution of the original elevation map, (p, p) denotes the resolution of each patch, and N is the resulting number of patches, which also serves as the effective input sequence length for the Transformer. The Transformer maintains a constant latent vector size D across all its layers. Therefore, we flatten the patches and map them to D dimensions using a trainable linear projection (see Equation 9). We refer to the output of this projection as the patch embeddings, the flatten process is described in Fig. 2.

To preserve positional information, positional embeddings are incorporated into the patch embeddings. We employ conventional learnable one-dimensional positional embeddings. The resultant sequence of embedding vectors is then fed into the transformer encoder.

$$P = \text{reshape}(F_{\text{integ}}, N, p), \quad N = \frac{H \times W}{p^2} \quad (8)$$

$$E = \text{Linear}(P), \quad E \in \mathbb{R}^{N \times D} \quad (9)$$

$$E_{\text{input}} = E + E_{\text{pos}} \quad (10)$$

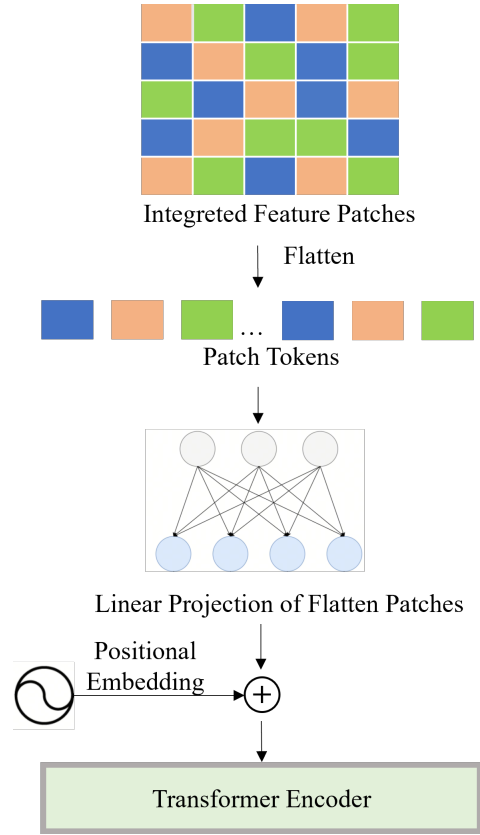


Fig. 2. The generation process of integrated features, the integrated feature map is firstly separated into several patches, after linear projection and positional embedding, the tensor generated is fed into transformer encoder.

3) *Encoder-Only Transformer Model:* The Transformer Encoder comprises 6 layers, each of which employs the self-attention mechanism to capture the intricate dependencies and contextual relationships among the embeddings. Formally, the self-attention operation within an encoder block can be expressed as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (11)$$

where Q , K , and V denote the query, key, and value matrices derived from the input embeddings, respectively, and d_k represents the dimensionality of the key vectors [13]. This mechanism enables the model to weigh the relevance of each patch relative to others, thereby enhancing the feature representations through contextual information. After processing through the Transformer Encoder, the enriched embeddings encapsulate comprehensive traversability information across the terrain. To translate these high-dimensional representations into actionable traversability assessments, we employ a multi-Layer perception (MLP). The MLP operates on each voxel's embedding to predict a traversability score within the range $[0, 10]$. The traversability prediction is mathematically defined as:

$$\hat{C} = \text{MLP}(E_{\text{output}}), \quad \hat{C} \in [0, 10] \quad (12)$$

where $\hat{C}$ represents the prediction result of traversability and E_{output} represents the output of the transformer encoder, respectively. The MLP comprises one or more fully connected layers with non-linear activation functions, enabling it to model complex relationships between the learned embeddings and the traversability outcomes.

The overall architecture ensures that the model effectively leverages both the spatial structure of the terrain and the dynamic vehicle stability information encapsulated within the embeddings. By integrating self-attention mechanisms and an MLP-based prediction layer, the framework achieves robust and accurate traversability assessments, facilitating reliable autonomous navigation in complex three-dimensional environments.

C. Training Details

The network is optimized by minimizing both reconstruction loss and traversability estimation loss. Reconstruction loss is calculated at each decoder layer using a weighted binary cross-entropy between the predicted pruning mask and the corresponding down-sampled target traversability cost. Traversability loss, on the other hand, is determined by the mean squared error between the predicted traversability scores and the target values, and it is only evaluated for correctly identified (true positive) points. The network is trained for 100,000 iterations with a batch size of 8, utilizing the ADAM optimizer with weight decay. The learning rate follows a 1cycle schedule, reaching a peak of 0.001. Additionally, during training, the correct pruning mask is enforced at each network layer to ensure training stability.

III. DATA COLLECTION AND SIMULATION

This section is organized as follows: section A. introduces the process of terrain generation and ground truth traversability evaluation, respectively. Section B. introduces the setup of comparative experiments conducted to compare the proposed method with a prevalent CNN-based approach [14], section C introduces the ablation experiments conducted to indicate the effect of the tricks proposed in the TETR model.

A. Ground Truth Traversability Data Collection

1) *Elevation map generation:* To create an extensive and varied terrain dataset that encompasses the range of complex three-dimensional environments, the foundational terrain layout is constructed using Perlin noise, producing a uniformly smooth surface. To develop a sophisticated setting populated with diverse static obstacles, numerous objects with diameters spanning from 0.1 meters to 6 meters are randomly placed within the map. A total of 100 objects are instantiated with randomized positions and orientations. The likelihood of each object being placed is inversely proportional to the size of its axis-aligned bounding box, resulting in a higher concentration of smaller objects. Additionally, each object's dimensions are varied by applying a random scaling factor between 0.5 and 1.5. In 90 of cases, the objects are positioned flush with the ground surface, while in the remaining 10, they are placed at

a random height ranging from 0 meters to 3 meters above the ground, creating overhanging structures.

2) *Physics-based Ground Truth Traversability Collection:* To evaluate the traversability cost effectively, we utilize the NVIDIA IsaacGym [15], a physics-based simulation platform that enables high-fidelity interaction with dynamically generated environments. The traversability is defined as the success rate of reaching the desired destination in the simulation platform.

For each sampled point of the elevation map, we determine whether the UGV can be placed without collisions, maintaining a standard joint configuration above that point. This assessment is performed for all heading angles α_k sampled at 10° increments, yielding 36 potential starting orientations per location. The UGV's pitch and roll orientations are synchronized with the underlying terrain mesh, and the base height is adjusted to ensure that all contact points are near the mesh surface [16]. For each voxel, matching the spatial resolution of the traversability cost tensor T , only one feasible UGV starting position per heading is further analyzed. To ensure the safety of the initialization position, a 2-second simulation of the UGV standing is conducted within a physics simulator using the current mobility policy. If the UGV remains stable and collision-free during this period, the pose is designated as a valid starting position [17]. Identifying reliable starting poses is crucial to minimize failures in traversability data collection caused by improper UGV initialization. For each validated standing pose, a total of $n_{\text{total}} = 10$ motion commands c_i are generated for the UGV, varying in steering angles and speeds. Specifically, the turning angles range from -20° to 20° in increments of 10° , and the speeds of the UGV is separated as creeping and cruising. Each command produces a UGV trajectory, which is evaluated for success based on criteria such as contact stability, base orientation, environmental collisions. Each command is executed by the UGV for 5 seconds. UGVs moving forward and sideways successfully reach their goals, whereas others fail to perform the intended motion commands due to environmental collisions. The overall traversability cost is calculated by dividing the number of successful attempts n_{suc} by the total number of attempts n_{total} , the ground truth traversability of each voxel is defined as [18]:

$$C = \frac{n_{\text{suc}}}{n_{\text{total}}} \quad (13)$$

B. Experiment and Result

For the training of our proposed method, we assembled 20 training terrain patches and 10 validation patches, in accordance with the procedure outlined in section III-A. The evaluation of each patch was performed using two NVIDIA A800-SXM4-80GB GPUs alongside 28 virtual CPU cores from an AMD EPYC 7763 processor. Table I presents the comparative results of the proposed method and the CNN-based baseline across the defined evaluation metrics.

Prediction Accuracy and MAE: The proposed TETR model achieves a significant improvement in prediction accuracy, attaining 93.1% compared to the CNN-based approach's

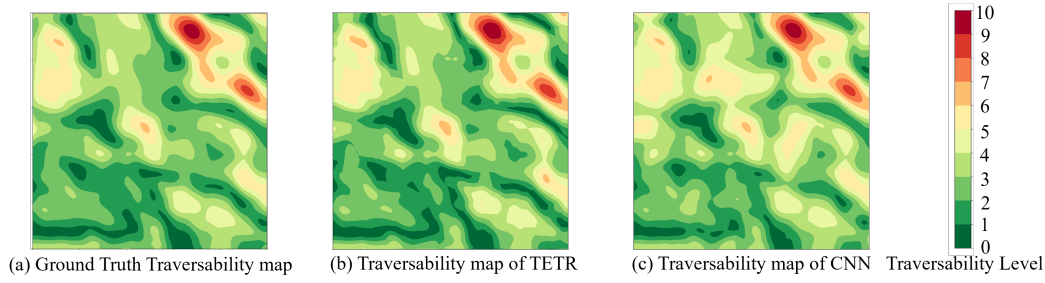


Fig. 3. Distribution of traversability map predicted by multiple methods, (a) represents the distribution evaluated by the physics engine, (b) represents the distribution evaluated by TETR model and (c) represents the distribution evaluated by CNN-based method, respectively. The color bar on the right represents the traversability level, where a higher level indicates that the current terrain is more difficult to traverse.

TABLE I
COMPARISON OF TETR AND CNN-BASED METHOD

Method	Accuracy (%)	MAE	Inference Time (ms)
CNN-Based	88.5	1.1	25.7
TETR	93.1	0.8	22.3

88.5%, which is increased by 5.2%. Additionally, the mean absolute error (MAE) is reduced to 0.8, which is reduced by 27.1%, demonstrating higher precision in traversability evaluation.

Computational Efficiency: The proposed TETR model exhibits a lower computational overhead with an average inference time of 22.3 ms per elevation map, compared to the CNN-based method's 25.7 ms, which is reduced by 16.9%. This reduction in inference time underscores the efficiency of our approach, making it more suitable for real-time autonomous navigation applications.

Traversability Cost Distribution: A comparison between the ground truth traversability map and those generated by the proposed TETR model and the CNN-based method is illustrated in Fig. 3. It is evident that the TETR-generated traversability distribution more closely aligns with the real traversability map compared to the CNN-based approach. Specifically, TETR effectively captures the nuanced variations in terrain, resulting in a smoother and more accurate representation of passable regions. In contrast, the CNN-based method exhibits scattered and discrete impassable areas, which deviate from the continuous nature of the actual traversability.

The superior performance of the transformer-based method can be attributed to its ability to capture long-range dependencies and contextual information through self-attention mechanisms. Comparing with CNN-based method, which primarily focus on local feature extraction, transformers can effectively integrate both spatial and dynamic vehicle stability information, leading to more accurate traversability assessments. Furthermore, the reduced computational overhead facilitates real-time implementation, which is critical for autonomous navigation systems operating in dynamic and unpredictable environments.

C. Ablation Studies

To further validate the effectiveness of key components in our transformer-based model, we conduct ablation studies by systematically removing or modifying these components and observing the impact on performance. The results, summarized in Table II, highlight the importance of each component in achieving high prediction accuracy and computational efficiency.

TABLE II
RESULTS OF ABLATION EXPERIMENTS

Ablation	Accuracy (%)	MAE	Inference Time (ms)
Full Model	93.1	0.8	22.3
-Positional Embedding	88.9	1.1	20.0
-Residual Connection	85.2	1.3	20.7

Impact of Positional Embeddings: Removing positional embeddings leads to a significant decrease in prediction accuracy from 93.1% to 88.9%, resulting in a 4.2% point reduction. Additionally, the MAE increases by 37.5%, rising from 0.8 to 1.1. Although the inference time slightly decreases by 10.3% from 22.3 ms to 20.0 ms, the substantial drop in accuracy and increase in MAE underscore the critical role of positional embeddings in capturing spatial dependencies within the model.

Impact of Residual Connections: Excluding residual connections further degrades the model's performance, with prediction accuracy dropping to 85.2%, a decrease of 7.9% points compared to the full model. The MAE increases by 62.5%, moving from 0.8 to 1.3. The inference time experiences a marginal reduction of 7.6%, decreasing from 22.3 ms to 20.7 ms. These changes highlight the importance of residual connections in maintaining high prediction accuracy and low error rates by effectively integrating original elevation features with dynamically encoded information.

IV. CONCLUSION

In this paper, we introduced TETR, a transformer-based model for traversability evaluation in autonomous navigation of unmanned ground vehicles operating in complex three-dimensional terrains. By integrating terrain features with vehicle kinematics through a principle-based encoding process and

utilizing a self-supervised training framework with a physics engine, TETR effectively captures both local and global contextual information, surpassing traditional CNN-based methods. Our experiments demonstrated that TETR achieves a prediction accuracy of 93.1%, marking a 5.2% improvement, while reducing computational overhead by 16.9%, thereby enabling real-time implementation. TETR offers a robust and efficient solution for traversability assessment, paving the way for safer and more reliable autonomous navigation in challenging terrains, with future work aimed at incorporating dynamic environmental factors and real-world validation. In future work, we plan to extend our model by incorporating dynamic environmental factors such as moving obstacles and varying weather conditions to enhance its robustness in real-world scenarios. Additionally, we aim to validate the model's performance with real-world experiments using actual UGVs in diverse terrains. Exploring the adaptation of the proposed approach to other types of vehicles and navigation tasks is also a promising direction for further research.

REFERENCES

- [1] C. Huang, C. Lv, P. Hang, and Y. Xing, "Toward safe and personalized autonomous driving: Decision-making and motion control with DPF and CDT techniques," *IEEE/ASME Transactions on Mechatronics*, vol. 26, no. 2, pp. 611–620, 2021.
- [2] T. Qie, W. Wang, C. Yang, Y. Li, Y. Zhang, W. Liu, and C. Xiang, "An improved model predictive control-based trajectory planning method for automated driving vehicles under uncertainty environments," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 4, pp. 3999–4015, Apr. 2022.
- [3] B. Paden, M. Cap, S. Z. Yong, D. Yershov, and E. Frazzoli, "A survey of motion planning and control techniques for self-driving urban vehicles," *IEEE Transactions on Intelligent Vehicles*, vol. 1, no. 1, pp. 33–55, Mar. 2016.
- [4] P. Papadakis, "Terrain traversability analysis methods for unmanned ground vehicles: A survey," *Engineering Applications of Artificial Intelligence*, vol. 26, pp. 1373–1385, Apr. 2013.
- [5] L. Wellhausen, A. Dosovitskiy, R. Ranftl, K. Walas, C. Cadena, and M. Hutter, "Where should I walk? Predicting terrain properties from images via self-supervised learning," *IEEE Robotics and Automation Letters*, vol. 4, no. 4, pp. 1509–1516, Apr. 2019.
- [6] R. O. Chavez-Garcia, J. Guzzi, L. M. Gambardella, and A. Giusti, "Learning ground traversability from simulations," *IEEE Robotics and Automation Letters*, vol. 3, no. 3, pp. 1695–1702, Jul. 2018.
- [7] J. Hu, Y. Hu, C. Lu, J. Gong, and H. Chen, "Integrated path planning for unmanned differential steering vehicles in off-road environment with 3D terrains and obstacles," *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 6, pp. 5562–5572, 2021.
- [8] J. Guzzi, R. O. Chavez-Garcia, M. Nava, L. M. Gambardella, and A. Giusti, "Path planning with local motion estimations," *IEEE Robotics and Automation Letters*, vol. 5, no. 2, pp. 2586–2593, Apr. 2020.
- [9] B. Graham, M. Engelcke, and L. van der Maaten, "3D semantic segmentation with submanifold sparse convolutional networks," in *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2018, pp. 1509–1516.
- [10] H. Li, W. Liu, C. Yang, W. Wang, T. Qie, and C. Xiang, "An optimization-based path planning approach for autonomous vehicles using the DynEFWA-artificial potential field," *IEEE Transactions on Intelligent Vehicles*, vol. 7, no. 2, pp. 263–272, 2021.
- [11] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 770–778.
- [12] A. Dosovitskiy, "An image is worth 16x16 words: Transformers for image recognition at scale," *arXiv preprint arXiv:2010.11929*, 2020.
- [13] A. Vaswani *et al.*, "Attention is all you need," in *Advances in Neural Information Processing Systems*, 2017, pp. 5998–6008.
- [14] G. G. Waibel, T. Löw, M. Nass, D. Howard, T. Bandyopadhyay, and P. V. K. Borges, "How rough is the path? terrain traversability estimation for local and global path planning," *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 9, pp. 16462–16473, 2022.
- [15] V. Makovychuk, L. Wawrzyniak, Y. Guo, M. Lu, K. Storey, M. Macklin, D. Hoeller, N. Rudin, A. Allshire, A. Handa, and G. State, "Isaac Gym: High performance GPU based physics simulation for robot learning," in *Neural Information Processing Systems*, Aug. 2021.
- [16] Y. Ji, Y. Tanaka, Y. Tamura, M. Kimura, A. Umemura, Y. Kaneshima, H. Murakami, A. Yamashita, and H. Asama, "Adaptive motion planning based on vehicle characteristics and regulations for off-road UGVs," *IEEE Transactions on Industrial Informatics*, vol. 15, no. 1, pp. 599–611, Jan. 2018.
- [17] C. Chen, J. Frey, P. Arm, and M. Hutter, "SMUG Planner: A safe multi-goal planner for mobile robots in challenging environments," *IEEE Robotics and Automation Letters*, 2023.
- [18] J. Frey, D. Hoeller, S. Khattak, and M. Hutter, "Locomotion policy guided traversability learning using volumetric representations of complex environments," in *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022, pp. 5722–5729.